28/01/2025
Evening

Aakash
Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1

[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. If $\sum_{r=1}^{13}\left\{\frac{1}{\sin\left(\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)}\right\}=a\sqrt{3}+b$, a, $b\in\mathbb{Z}$, then a^2+b^2 is equal to:

(1) 2

(2) 8

(3) 10

(4) 4

Answer (2)

Sol. $S=\sum_{r=1}^{13}\frac{\sin\left(\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)-\left(\frac{\pi}{4}+\left(\frac{r-1}{4}\right)\frac{\pi}{6}\right)\right)}{\sin\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)\sin\left(\frac{\pi}{4}+\left(\frac{(r-1)\pi}{6}\right)\right)}\times\frac{1}{\sin\left(\frac{\pi}{6}\right)}$

$\Rightarrow S=2\sum_{r=1}^{13}\frac{\cos\left(\frac{\pi}{4}+\frac{(r-1)\pi}{6}\right)}{\sin\left(\frac{\pi}{4}+\frac{(r-1)\pi}{6}\right)}-\frac{\cos\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)}{\sin\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)}$

$\Rightarrow S=2\sum_{r=1}^{13}\cot\left(\frac{\pi}{4}+\frac{(r-1)\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)$

$=2\left(\cot\left(\frac{\pi}{4}+\frac{0\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{\pi}{6}\right)\right)$

$+\cot\left(\frac{\pi}{4}+\frac{\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{2\pi}{6}\right)$ .....

$+\cot\left(\frac{\pi}{4}+\frac{12\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{13\pi}{6}\right)$

$=2\left(\cot\left(\frac{\pi}{4}\right)-\cot\left(\frac{\pi}{4}+\frac{13\pi}{6}\right)\right)=2\sqrt{3}-2$

2. Bag B_1 contains 6 white and 4 blue balls, Bag B_2 contains 4 white and 6 blue balls, and Bag B_3 contains 5 white and 5 blue balls. One of the bags is selected at random and a ball is drawn from it. If the ball is white, then the probability, that the ball is drawn from Bag B_2, is

(1) $\frac{2}{5}$ (2) $\frac{1}{3}$

(3) $\frac{2}{3}$ (4) $\frac{4}{15}$

Answer (4)

Sol. $P\left(\frac{B_2}{w}\right)=\frac{P(B_2)\cdot P\left(\frac{w}{B_2}\right)}{P(w)}$

$P(Bi)=\frac{1}{3},\ i=1,2,3$

$P\left(\frac{w}{B_1}\right)=\frac{6}{10}, P\left(\frac{w}{B_2}\right)=\frac{4}{10}, P\left(\frac{w}{B_3}=\frac{5}{10}\right)$

$P(w)=\sum_{i=1}^{3}P(B_i)P\left(\frac{w}{B_i}\right)=\frac{1}{3}\left(\frac{6}{10}+\frac{4}{10}+\frac{5}{10}\right)$

$=\frac{5}{10}=\frac{1}{2}$

$P\left(\frac{B_2}{w}\right)=\frac{\frac{1}{3}\cdot\frac{4}{10}}{\frac{1}{2}}=\frac{8}{30}=\frac{4}{15}$

3. Let S be the set of all the words that can be formed by arranging all the letters of the word GARDEN. From the set S, one word is selected at random. The probability that the selected word will **NOT** have vowels in alphabetical order is:

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $\frac{1}{4}$ (4) $\frac{2}{3}$

Answer (1)



Sol. V : vowel in alphabet order

$$P(\bar{v}) = 1 - P(v)$$

$$= 1 - \frac{\left(\frac{6!}{2!}\right)}{6!} = 1 - \frac{1}{2} = \frac{1}{2}$$

4. Let the coefficients of three consecutive terms T_r, T_{r+1} and T_{r+2} in the binomial expansion of $(a+b)^{12}$ be in a G.P. and let p be the number of all possible values of r. Let q be the sum of all rational terms in the binomial expansion $\left(\sqrt[4]{3}+\sqrt[3]{4}\right)^{12}$. Then $p + q$ is equal to:

(1) 287 (2) 295

(3) 299 (4) 283

Answer (4)

Sol. Coefficient of

$$T_r, T_{r+1}, T_{r+2} \rightarrow GP$$

$$\Rightarrow \left({}^{12}C_r\right)^2 = {}^{12}C_{r-1} \cdot {}^{12}C_{r+1}$$

but no three consecutive binomial coefficient are in GP

$$\Rightarrow P = 0$$

Now for $\left(3^{1/4}+4^{1/3}\right)^{12}$, $T_{r+1} = {}^{12}C_r (4)^{K/3} (3)^{\frac{12-K}{4}}$

for rational terms K =0, 12

sum of rational terms

$= {}^{12}C_0 4^0 \cdot 3^3 + {}^{12}C_{12} \cdot 4^4 \cdot 3^0$

$= 27 + 256 = 283 = q$

$\therefore p + q = 283$

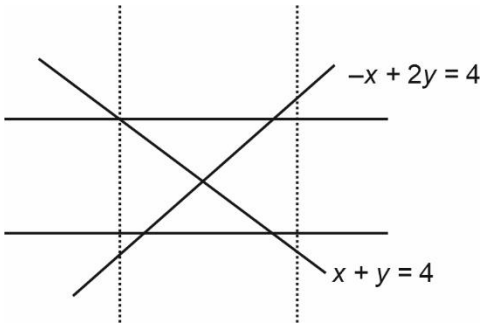
5. Two equal sides of an isosceles triangle are along $-x + 2y = 4$ and $x + y = 4$. If m is the slope of its third side, then the sum, of all possible distinct values of m, is:

(1) $-2\sqrt{10}$ (2) 6

(3) 12 (4) –6

Answer (2)

Sol.



Slope of the third side = slope of the perpendicular bisector of given lines

$$h: \frac{-x+2y-4}{\sqrt{5}} = \pm\frac{x+y-4}{\sqrt{2}}$$

$$h_1: \sqrt{2}(-x+2y-4) = \sqrt{5}(x+y-4)$$

$$h_2: \sqrt{2}(-x+2y-4) = -\sqrt{5}(x+y-4)$$

$$M_{L_1}: -\left[\frac{\sqrt{5}+\sqrt{2}}{\sqrt{5}-2\sqrt{2}}\right]$$

$$M_{L_2}: -\left[\frac{\sqrt{5}-\sqrt{2}}{\sqrt{5}+2\sqrt{2}}\right]$$

$$M_{L_1} + M_{L_2} = -\left[\frac{\sqrt{5}+\sqrt{2}}{\sqrt{5}-2\sqrt{2}} + \frac{\sqrt{5}-\sqrt{2}}{\sqrt{5}+2\sqrt{2}}\right]$$

$$= -\left[\frac{\left(\sqrt{5}+\sqrt{2}\right)\left(\sqrt{5}+2\sqrt{2}\right)+\left(\sqrt{5}-\sqrt{2}\right)\left(\sqrt{5}-2\sqrt{2}\right)}{-3}\right]$$

$= 6$

6. Let $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & -2 \\ 0 & 1 \end{bmatrix}$ and $P = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, $\theta > 0$.

If $B = PAP^T$, $C = P^T B^{10} P$ and the sum of the diagonal elements of C is $\frac{m}{n}$, where $gcd(m, n) = 1$, then $m + n$ is:

(1) 127 (2) 65

(3) 2049 (4) 258

Answer (2)



Sol. $A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -2 \\ 0 & 1 \end{pmatrix}$

$P = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

$PP^T = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

$C = P^T B B^9 P$

$= P^T PA P^T B^9 P$

$= AP^T PA P^T B^8 P$

$= A^2 P^T B^8 P$

$\vdots$

$= A^{10}$

Sum of diagonal elements of $A^{10} = \left(\frac{1}{\sqrt{2}}\right)^{10} + 1$

$= \frac{1}{32} + 1 = \frac{33}{32} = \frac{m}{n}$

So, $m + n = 32 + 33$

$= 65$

7. If $\alpha + i\beta$ and $\gamma + i\delta$ are the roots of $x^2 - (3 - 2i)x - (2i - 2) = 0$, $i = \sqrt{-1}$, then $\alpha\gamma + \beta\delta$ is equal to:

(1) 6 (2) –2

(3) 2 (4) –6

Answer (3)

Sol. $x^2 - (3 - 2i)x - (2i - 2) = 0$ (roots p, q)

$p + q = 3 - 2i = 1 + (2 - 2i)$

$pq = 2 - 2i = 1(2 - 2i)$

$\Rightarrow \quad p = 1 + 0i,\ q = 2 - 2i$

Comparing $\alpha\gamma = 2 \Rightarrow \alpha\gamma + \beta\delta = 2$

$\beta\delta = 0(-2)$

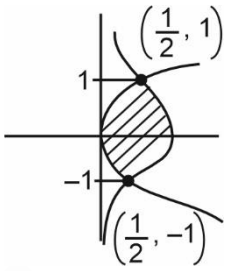
8. The area of the region bounded by the curves $x(1 + y^2) = 1$ and $y^2 = 2x$ is:

(1) $\frac{\pi}{2} - \frac{1}{3}$ (2) $\frac{1}{2}\left(\frac{\pi}{2} - \frac{1}{3}\right)$

(3) $2\left(\frac{\pi}{2} - \frac{1}{3}\right)$ (4) $\frac{\pi}{4} - \frac{1}{3}$

Answer (1)

Sol.



$x = \frac{1}{1 + y^2},\ x = \frac{y^2}{2}$

Solving, $\frac{1}{1 + y^2} = \frac{y^2}{2} \Rightarrow y = \pm 1$

Area $= \int_{-1}^{1}\left(\frac{1}{1 + y^2} - \frac{y^2}{2}\right)dy = \tan^{-1} y - \frac{y^3}{6}\Big|_{-1}^{1} = \frac{\pi}{2} - \frac{1}{3}$

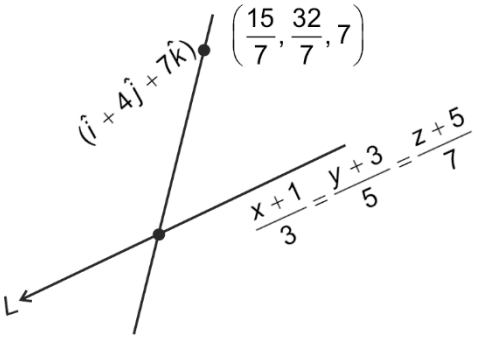
9. The square of the distance of the point $\left(\frac{15}{7}, \frac{32}{7}, 7\right)$ from the line $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ in the direction of vector $\hat{i} + 4\hat{j} + 7\hat{k}$ is

(1) 66 (2) 54

(3) 41 (4) 44

Answer (1)

Sol.



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$\Rightarrow \left(\lambda+\frac{15}{7}, 4\lambda+\frac{32}{7}, 7\lambda+7\right)$ for some λ lie on L.

$\Rightarrow \frac{\lambda+15/7+1}{3}=\frac{7\lambda+7+5}{7}$

$\Rightarrow \lambda=-1 \Rightarrow PQ=\sqrt{66}$

$\Rightarrow PQ^2 = 66$

10. Let $f: \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $f(2)=1$. If $F(x)=xf(x)$ for all $x\in\mathbf{R}$, $\int_0^2 xF'(x)dx=6$ and $\int_0^2 x^2F''(x)dx=40$, then $F'(2)+\int_0^2 F(x)dx$ is equal to:

(1) 15 (2) 9

(3) 13 (4) 11

Answer (4)

Sol. $\int_0^2 xF'(x)=6$

$\left[F'(x)\cdot\frac{x^2}{2}\right]_0^2-\int_0^2 F''(x)\cdot\frac{x^2}{2}dx=6$

$2F'(2) = 26 \Rightarrow F'(2) = 13$

Given : $F(x) = xf(x)$

$F'(x) = xf'(x) + f(x)$

Put $x = 2$

$F'(2) = 2f'(2) + f(2)$

$f'(2) = 6$

As $\int_0^2 xF'(x)=6$

$\Rightarrow [xF(x)]_0^2-\int_0^2 F(x)dx=6$

$\Rightarrow \int_0^2 F(x)dx=-2$

$\therefore F'(2)+\int_0^2 F(x)dx = 13-2=11$

11. If the components of $\vec{a}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ along and perpendicular to $\vec{b}=3\hat{i}+\hat{j}-\hat{k}$ respectively, are $\frac{16}{11}\left(3\hat{i}+\hat{j}-\hat{k}\right)$ and $\frac{1}{11}\left(-4\hat{i}-5\hat{j}-17\hat{k}\right)$, then $\alpha^2+\beta^2+\gamma^2$ is equal to:

(1) 23 (2) 18

(3) 26 (4) 16

Answer (3)

Sol. Component of $\vec{a}$ along $\vec{b}$ is $\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\right)\vec{b}$

$\frac{(\alpha,\beta,\gamma)\cdot(3,1,-1)}{\sqrt{11}}(3\hat{i}+\hat{j}-\hat{k})$

$\frac{(3\alpha+\beta-\gamma)}{\sqrt{11}}(3\hat{i}+\hat{j}-\hat{k})$

$=\frac{16}{11}(3\hat{i}+\hat{j}-\hat{k})$

$\vec{a}-\left(\vec{a}\cdot\vec{b}\right)\cdot b=\overrightarrow{v_2}$

$\vec{a}=\overrightarrow{v_1}+\overrightarrow{v_2}$

$\vec{a}=\frac{(44,11,-33)}{11}$

$\therefore \alpha = 4, \beta = 1, \gamma = -3.$

12. If $f(x)=\int\frac{1}{x^{1/4}(1+x^{1/4})}dx$, $f(0)=-6$, then $f(1)$ is equal to :

(1) $\log_e 2 + 2$ (2) $2 - \log_e 2$

(3) $4(\log_e 2 - 2)$ (4) $4(\log_e 2 + 2)$

Answer (3)

Sol. Put $x^{1/4}=t \Rightarrow dx=4t^3dt$

$\int\frac{4t^3dt}{t(t+1)}=4\int\left(\frac{t^2-1}{t+1}+\frac{1}{t+1}\right)dt$

$f(x)=4\left[\frac{x^{1/2}}{2}-x^{1/4}+\ln|x^{1/4}+1|\right]+C$



$f(0) = -6$

$\Rightarrow C = -6$

$\Rightarrow f(1) = 4(\log_e 2 - 2)$

13. If A and B are the points of intersection of the circle $x^2 + y^2 - 8x = 0$ and the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ and a point P moves on the line $2x - 3y + 4 = 0$, then the centroid of ΔPAB lies on the line:

(1) $4x - 9y = 12$ (2) $6x - 9y = 20$

(3) $9x - 9y = 32$ (4) $x + 9y = 36$

Answer (2)

Sol. C : $x^2 + y^2 - 8x = 0$ H : $\frac{x^2}{9} - \frac{y^2}{4} = 1$

By solving $\frac{x^2}{9} - \left(\frac{8x - x^2}{4}\right) = 1$

$4x^2 - 72x + 9x^2 = 36$

$\Rightarrow 13x^2 - 72x - 36 = 0$

$\Rightarrow 13x^2 - 78x + 6x - 36 = 0$

$\Rightarrow 13x(x - 6) + 6(x - 6) = 0$

$\Rightarrow x = 6$ or $\boxed{-\frac{13}{6}}$ × neglected

$\Rightarrow y^2 = 8(6) - (6)^2$

$\Rightarrow y = \pm\sqrt{12}$

So, points A and B are $(6, \sqrt{12}), (6, -\sqrt{12})$

$P(h, \frac{2h+4}{3})$

Centroid of ΔPAB is $\left(\frac{12+h}{3}, \frac{2h+4}{9}\right)$

By options this centroid lies on the live $6x - 9y = 20$

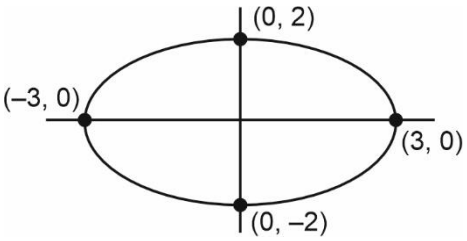
14. If the midpoint of a chord of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is $\left(\sqrt{2}, \frac{4}{3}\right)$, and the length of the chord is $\frac{2\sqrt{\alpha}}{3}$, then α is

(1) 26 (2) 22

(3) 20 (4) 18

Answer (2)

Sol.



$E : \frac{x^2}{9} + \frac{y^2}{4} = 1$

$T = S_1$

$\Rightarrow \frac{\sqrt{2}x}{9} + \frac{1}{4}\left(\frac{4}{3}y\right) - 1 = \frac{2}{9} + \frac{16}{9(4)} - 1$

$\frac{\sqrt{2}x}{9} + \frac{y}{3} = \frac{2}{9} + \frac{4}{9}$

$\frac{\sqrt{2}x}{9} + \frac{y}{3} = \frac{2}{3} \Rightarrow \boxed{\sqrt{2}x + 3y = 6}$

Now point of intersection of chord and ellipse is

$\frac{(6-3y)^2}{18} + \frac{y^2}{4} = 1$

$\frac{(2-y)^2}{2} + \frac{y^2}{4} = 1$

$2(4 + y^2 - 4y) + y^2 = 4$

$\Rightarrow 3y^2 - 8y + 4 = 0$

$\Rightarrow y = 2, \frac{2}{3}$

So, points are (0, 2) are $\left(2\sqrt{2}, \frac{2}{3}\right)$

Length of chord $= \sqrt{(2\sqrt{2})^2 + \left(\frac{2}{3} - 2\right)^2}$



$= \sqrt{8+\frac{16}{9}}$

$= \frac{\sqrt{88}}{3} = \frac{2\sqrt{22}}{3}$

On comparing $\alpha = 22$

15. Let $[x]$ denote the greatest integer less than or equal to x. Then the domain of $f(x) = \sec^{-1}(2[x]+1)$ is:

(1) $(-\infty,-1]\cup[1,\infty)$ (2) $(-\infty,-1]\cup[0,\infty)$

(3) $(-\infty,\infty)$ (4) $(-\infty,\infty)-\{0\}$

Answer (3)

Sol. $f(x) = \sec^{-1}(2[x]+1)$

$\Rightarrow 2[x]+1 \geq 1$ or $2[x]+1 \leq -1$

$\Rightarrow 2[x] \geq 0$ or $2[x] \leq -2$

$\Rightarrow [x] \geq 0$ or $[x] \leq -1$

$\Rightarrow x \geq 0$ or $x \leq 0$

Domain of $f(x)$ is $(-\infty, \infty)$

16. Let $f : [0, 3] \to A$ be defined by $f(x) = 2x^3 - 15x^2 + 36x + 7$ and $g : [0, \infty) \to B$ be defined by $g(x) = \frac{x^{2025}}{x^{2025}+1}$. If both the functions are onto and $S = \{x \in \mathbf{Z} : x \in A \text{ or } x \in B\}$, then $n(S)$ is equal to :

(1) 36 (2) 30

(3) 29 (4) 31

Answer (2)

Sol. $f(x) = 2x^3 - 15x^2 + 36x + 7$

$f'(x) = 6x^3 - 30x + 36 = 0$

$\Rightarrow x^2 - 5x + 6 = 0$

$\therefore x = 1, 5$

$f(0) = 7$, $f(2) = 35$, $f(3) = 34$

$A = [7, 35]$

$g(x) = \frac{x^{2025}}{1+x^{2025}}$

$B = [0, 1)$

$\therefore S = [0, 7, 8, 9 \ldots.35]$

Number of elements = 30

17. Let A, B, C be three points in xy-plane, whose position vector are given by $\sqrt{3}\hat{i}+\hat{j}$, $\hat{i}+\sqrt{3}\hat{j}$ and $a\hat{i}+(1-a)\hat{j}$ respectively with respect to the origin O. If the distance of the point C from the line bisecting the angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is $\frac{9}{\sqrt{2}}$, then the sum of all the possible values of a is :

(1) 2 (2) 0

(3) 1 (4) $\frac{9}{2}$

Answer (3)

Sol. Equation of line in the internal bisector of $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is $\left(\sqrt{3}+1\right)\hat{i}+\left(\sqrt{3}+1\right)\hat{j}$

$\Rightarrow$ line will be $y = x \Rightarrow x - y = 0$

$D = \left|\frac{a-(1-a)}{\sqrt{a^2+(1-a)^2}}\right| = \frac{9}{\sqrt{2}}$

$(2a-1)^2 = \frac{81}{2}(a^2 + (1-a)^2)$

$\Rightarrow 2(4a^2 - 4a + 1) = 81a^2 + 81a^2 - 162a - 81$

$\Rightarrow 162a^2 - 162a + 81 - 8a^2 + 8a - 2 = 0$

$\Rightarrow 154a^2 - 154a + 79 = 0$

Sum of values $= -\frac{(-154)}{154} = 1$

18. Let f be real valued continuous function defined on the positive real axis such that $g(x) = \int_0^x tf(t)dt$. If $g(x^3) = x^6 + x^7$, then value of $\sum_{r=1}^{15} f(r^3)$ is :

(1) 270 (2) 310

(3) 340 (4) 320

Answer (2)

Sol. $g(x) = x^2 + x^{7/3}$



$$g'(x) = 2x + \frac{7}{3}x^{4/3}$$

$$f(x) = \frac{g'(x)}{x}$$

$$f(x) = 2 + \frac{7}{3}x^{1/3}$$

$$f(r^3) = 2 + \frac{7}{3}r$$

$$\sum_{r=1}^{15}\left(2 + \frac{7}{3}r\right) = 2(15) + \frac{7}{3}\left(\frac{15(16)}{2}\right)$$

$$= 310$$

19. For positive integers n, if $4a_n = (n^2 + 5n + 6)$ and $s_n = \sum_{k=1}^{n}\left(\frac{1}{a_k}\right)$, then the value of $507\ S_{2025}$ is:

(1) 135

(2) 1350

(3) 675

(4) 540

Answer (3)

Sol. $$S_n = \sum_{k=1}^{n}\frac{4}{K^2 + 5k + 6}$$

$$= \sum_{k=1}^{n}\frac{4}{(K+2)(K+3)} = 4\sum_{K=1}^{n}\left(\frac{1}{K+2} - \frac{1}{K+3}\right)$$

$$= 4\left[\frac{1}{3} - \frac{1}{4}\right]$$

$$= 4\left[\frac{1}{4} - \frac{1}{5}\right]$$

$$= 4\left[\frac{1}{n+2} - \frac{1}{n+3}\right]$$

$$S_n = 4\left[\frac{1}{3} - \frac{1}{n+3}\right]$$

$$S_{2025} = 4\left[\frac{1}{3} - \frac{1}{2028}\right]$$

$$S_{2025} = 4\left[\frac{675}{2028}\right]$$

$$507\ S_{2025} = 675$$

20. Let $f : R - \{0\} \to (-\infty, 1)$ be a polynomial of degree 2, satisfying $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$. If $f(K) = -2K$, then the sum of squares of all possible values of K is :

(1) 6

(2) 7

(3) 9

(4) 1

Answer (1)

Sol. Let $f(x) = ax^2 + bx + c$

Putting in given function at equation

and comparing coefficients given

$c = 1$, $a = \pm 1$

Hence $f(x) = 1 - x^2$

$f(k) = -2k \Rightarrow k^2 - 2k - 1 = 0 \begin{matrix}\alpha \\ \beta\end{matrix}$

$\Rightarrow \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= 4 - 2(-1) = 6$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. The interior angles of a polygon with n sides, are in an A.P, with common difference 6°. If the largest interior angle of the polygon is 219°, then n is equal to _____.

Answer (20)

Sol. $$\frac{n}{2}[2a + (n-1)6] = (n-2)180°$$



and $an + 3n^2 - 3n = 3n\,(n-2)180°$ ....(i)

$\therefore$ Given $a + (n-1)6° = 219°$

$\Rightarrow$ $a = 225° - 6n°$

Putting value of a in (i)

We get $(225 - 6n^2) + 3n^2 - 3n = 180n - 360°$

$\Rightarrow$ $2n^2 - 42n - 360 = 0$

$\Rightarrow$ $n^2 - 14n - 120 = 0$

$\Rightarrow$ $(n-20)\,(n+6) = 0$

$\Rightarrow$ $n = 20, -6$ (Rejected)

$\therefore$ $n = 20$

22. If $f(x) = \lim\limits_{n\to\infty} \sum\limits_{r=0}^{n} \left(\dfrac{\tan\left(x/2^{r+1}\right) + \tan^3\left(x/2^{r+1}\right)}{1 - \tan^2\left(x/2^{r+1}\right)} \right)$.

Then $\lim\limits_{x\to 0} \dfrac{e^x - e^{f(x)}}{(x - f(x))}$ is equal to _____.

Answer (01)

Sol. $f(x) = \lim\limits_{x\to\infty} \left(\dfrac{\tan\left(x/2^{r+1}\right) + \tan^3\left(x/2^{r+1}\right)}{1 - \tan^2\left(x/2^{r+1}\right)} \right)$

$= \lim\limits_{x\to\infty} \dfrac{\tan\left(\dfrac{x}{2^{r+1}}\right)}{\cos\left(\dfrac{x}{2^r}\right)}$

$= \lim\limits_{x\to\infty} \dfrac{\sin\left(x/2^{r+1}\right)}{\cos\left(\dfrac{x}{2^{r+1}}\right)\cos\left(\dfrac{x}{2^r}\right)}$

$= \lim\limits_{x\to\infty} \dfrac{\sin\left(\dfrac{x}{2^r} - \dfrac{x}{2^{r+1}}\right)}{\cos\left(\dfrac{x}{2^{r+1}}\right)\cos\left(\dfrac{x}{2^r}\right)}$

$= \lim\limits_{x\to\infty} \tan\left(\dfrac{x}{2^r}\right) - \tan\left(\dfrac{x}{2^{r+1}}\right)$

From condition given question

$\therefore$ $\lim\limits_{n\to\infty} \sum\limits_{r=0}^{n} \left[\tan\left(\dfrac{x}{2^r}\right) - \tan\left(\dfrac{x}{2^{r+1}}\right) \right] = \tan x$

$\therefore$ $\lim\limits_{x\to 0} \left(\dfrac{e^x - e^{\tan x}}{x - \tan x} \right)$

$\Rightarrow$ $\lim\limits_{x\to 0} e^{\tan x} \left(\dfrac{e^{x-\tan x} - 1}{x - \tan x} \right)$

$\Rightarrow$ $\lim\limits_{x\to 0} e^{\tan x} \lim\limits_{x\to 0} \left(\dfrac{e^{x-\tan x} - 1}{x - \tan x} \right)$

$\Rightarrow$ $1 \cdot 1 \left(\because \lim\limits_{x\to 0} \dfrac{e^{x-1}}{x} = 1 \right)$

$= 1$

23. If $y = y(x)$ is the solution of the differential equation,

$\sqrt{4 - x^2}\,\dfrac{dy}{dx} = \left(\left(\sin^{-1}\left(\dfrac{x}{2}\right) \right)^2 - y \right) \sin^{-1}\left(\dfrac{x}{2}\right), -2 \le x \le 2, y(2)$

$= \dfrac{\pi^2 - 8}{4}$, then $y^2(0)$ is equal to _____.

Answer (4)

Sol. $\dfrac{dy}{dx} + \dfrac{\left(\sin^{-1}\dfrac{x}{2}\right)}{\sqrt{4 - x^2}} y = \dfrac{\left(\sin^{-1}\dfrac{x}{2}\right)^3}{\sqrt{4 - x^2}}$

$y\, e^{\left(\frac{\sin^{-1}\frac{x}{2}}{2}\right)^2} = \int \dfrac{\left(\sin^{-1}\dfrac{x}{2}\right)^3}{4 - x^2} e^{\frac{\left(\sin^{-1}\frac{x}{2}\right)^2}{2}}\, dx$

$y = \left(\sin^{-1}\dfrac{x}{2}\right)^2 - 2 + c \cdot e^{\frac{-\left(\sin^{-1}\frac{x}{2}\right)^2}{2}}$

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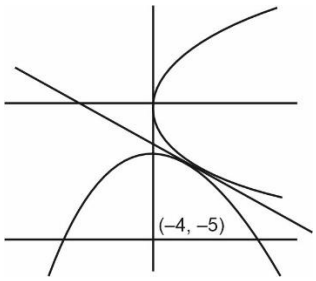
$$y(2) = \frac{\pi^2}{4} - 2 \Rightarrow c = 0$$

$$y(0) = -2$$

24. Let A and B be the points of intersection of the line $y + 5 = 0$ and the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If d denotes the distance between A and B, and a denotes the area of ΔSAB, where S is the focus of the parabola $y^2 = 4x$, then the value of $(a + d)$ is _____.

Answer (14)

Sol.



To find image of $P(t^2, 2t)$

$$\frac{x - t^2}{1} = \frac{y - 2t}{1} = \frac{-2(t^2 + 2t + 4)}{1^2 + 1^2} = -(t+1)^2 - 3$$

$x = t^2 - (t + 1)^2 - 3 = -2t - 4$

$y = 2t - (t + 1)^2 - 3 = -t^2 - 4$

$$t = \frac{-x - 4}{2}$$

$$\Rightarrow \quad y + 4 = -\left(\frac{-x-4}{2}\right)^2$$

$$\Rightarrow \quad (y+4) = -\frac{(x+4)^2}{4}$$

$\Rightarrow x^2 = -4y$

$\Rightarrow$ Focus $(-4, -5)$

Also, $y = -5$ intersect

$\therefore \quad (-4)(-1) = (x + 4)^2$

$4 = (x + 4)^2$

$x + 4 = \pm 2$

$x = -2, -6$

$\Rightarrow \quad d = 4$

$$a = \frac{1}{2}\begin{vmatrix} 1 & 0 & 1 \\ -2 & -5 & 1 \\ -6 & -5 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\left|[1(-5+5) + 1(10-30)]\right|$$

$$= \frac{1}{2}(20)$$

$a = 10$

$\therefore \quad a + d = 14$

25. The number of natural numbers, between 212 and 999, such that sum of their digits is 15, is _____.

Answer (64)

Sol. Let the number be

$2ab$, $a + b = 13$

$\Rightarrow \quad a, b \in \{0, 9\}$

$\Rightarrow$ 6 numbers $\{(9, 4), (8, 5) \ldots (4,9)\}$

Similarly, for $3ab$, $a + b = 12 \Rightarrow$ 7 numbers

For $4ab$, $a + b = 11 \Rightarrow$ Numbers

For $5ab$, $a + b = 10 \Rightarrow$ 9 numbers

For $6ab$, $a + b = 9 \Rightarrow$ 10 numbers

For $7ab$, $a + b = 8 \Rightarrow$ 9 numbers

For $8ab$, $a + b = 7 \Rightarrow$ 8 numbers

For $9ab$, $a + b = 6 \Rightarrow$ 7 numbers

$\therefore$ Total ways = 64.



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. A body of mass 4 kg is placed on a plane at a point P having coordinates (3, 4) m. Under the action of force $\vec{F}=\left(2\hat{i}+3\hat{j}\right)$ N, it moves to a new point Q having coordinates (6, 10) m in 4 sec. The average power and instantaneous power at the end of 4 sec are in the ratio of

(1) 1 : 2 (2) 6 : 13

(3) 4 : 3 (4) 13 : 6

Answer (2)

Sol. $\vec{F}=\left(2\hat{i}+3\hat{j}\right)$ N ; displacement $\vec{S}=\left(3\hat{i}+6\hat{j}\right)$ m

Acceleration, $\vec{a}=\frac{\vec{F}}{m}=\left(\frac{2\hat{i}+3\hat{j}}{4}\right)$m/s^2

$$\vec{S}=\vec{u}t+\frac{1}{2}\vec{a}t^2$$

$$\Rightarrow \quad 3\hat{i}+6\hat{j}=4\vec{u}+\frac{1}{2}\left(\frac{2\hat{i}+3\hat{j}}{4}\right)\cdot 16=4\vec{u}+4\hat{i}+6\hat{j}$$

or $\vec{u}=\frac{1}{4}\left(3\hat{i}+6\hat{j}-4\hat{i}-6\hat{j}\right)$

$$\vec{u}=-\frac{1}{4}\hat{i}$$

Average Power $=\vec{F}\cdot\frac{\vec{S}}{t}$

$$=\frac{\left(2\hat{i}+3\hat{j}\right)\cdot\left(3\hat{i}+6\hat{j}\right)}{4}\text{ watt}$$

= 6 watt

Instantaneous Power $=\vec{F}\cdot\vec{v}$

$$=\vec{F}\cdot\left(\vec{u}+\vec{a}t\right)$$

$$=\left(2\hat{i}+3\hat{j}\right)\cdot\left(-\frac{1}{4}\hat{i}+2\hat{i}+3\hat{j}\right)$$

$$=\left(2\hat{i}+3\hat{j}\right)\cdot\left(+\frac{7}{4}\hat{i}+3\hat{j}\right)$$

= (3.5 + 9) watt

= 12.5 watt

27. A 400 g solid cube having an edge of length 10 cm floats in water. How much volume of the cube is outside the water?

(Given : density of water = 1000 kg m^{-3})

(1) 600 cm^3

(2) 400 cm^3

(3) 4000 cm^3

(4) 1400 cm^3

Answer (1)

Sol. Volume of cube inside water

$$=\left(\frac{\text{Density of cube}}{\text{Density of water}}\right)\times\text{Volume of cube}$$

$$=\frac{\text{Mass of cube}}{\text{Density of water}}$$

$$=\frac{400\text{ gm}}{1\text{ gm/cm}^3}$$

= 400 cm^3

Volume of cube outside water

= Volume of cube – Volume of cube inside water

= 1000 cm^3 – 400 cm^3

= 600 cm^3

